Geometric Invariants of Grade-Resolved Fields: Hodograph and Egregium Readings of Stochastic Time Series

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Abstract

Hamilton (1847) showed that under inverse-square gravity, the hodograph—the curve traced by the velocity vector—is a circle of constant curvature, and concluded that the velocity channel admits an autonomous geometric description. Gauss (1827) proved that the Gaussian curvature of a surface is an *intrinsic* invariant, independent of the ambient embedding. Both results are exact statements about deterministic dynamics. We instantiate both readings on the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ of a time series, exhibiting a canonical structure: the geometric invariants are *exact on deterministic dynamics* (zero variance, exact analytical predictions), *asymptotically valid on stochastic dynamics* as the noise scale tends to zero, and possess a *detectability threshold* $\sigma^\star$ specific to each process—the noise capacity of the deterministic signature. The *hodograph reading* treats the bivector trajectory $B(t) = \mathbf{q}(t-1) \wedge \mathbf{q}(t)$ as a curve in $\mathbb{R}^3$ and computes its discrete Frenet-Serret curvature $\kappa(t)$; the *Egregium reading* treats the lagged scalar surface $s(t,\tau) = \mathbf{q}(t) \cdot \mathbf{q}(t+\tau)$ as a 2-D surface and computes its Gaussian curvature $K(t,\tau)$. Empirically: for deterministic processes (sine, polynomial, logistic chaos, Hénon), the geometric invariants are computed exactly. Detectability thresholds: $\sigma^\star \approx 0.3$ for logistic chaos, $\sigma^\star \approx 1.0$ for Hénon; periodic and polynomial-trend processes retain their signature at $\sigma > 3$. For stochastic processes (AR, GARCH, fBm), the asymptotic version of the framework yields t-statistics up to $+24$ (hodograph) and $+91$ (Egregium) for DGP discrimination. The framework is in this sense canonical in the manner of mathematical physics: an exact theory of deterministic dynamics, with stochastic dynamics as asymptotic perturbation and process-specific detectability thresholds as additional invariants.

Keywords. Geometric Signal Dynamics, Clifford algebra, hodograph, Theorema Egregium, kinematic embedding, Frenet-Serret, time series geometry.

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1 Introduction

1.1 Two ancestors, one programme

Two readings of geometric invariance, separated by twenty years and authored by Gauss and Hamilton, share a programmatic intuition that becomes operational in the empirical-stochastic context once one has the kinematic embedding of a time series. Gauss's *Disquisitiones Generales circa Superficies Curvas* (1827) proved that the Gaussian curvature K of a surface in $\mathbb{R}^3$ depends only on the *intrinsic* metric of the surface, not on its ambient embedding. Two surfaces that are *isometric* as Riemannian manifolds have identical K everywhere, even if they sit in $\mathbb{R}^3$ very differently. The result—Theorema Egregium—articulates a principle: certain geometric quantities are properties of the object itself, not of the particular coordinates one has chosen to describe it.

Hamilton's *The Hodograph, or a New Method of Expressing in Symbolical Language the Newtonian Law of Attraction* (1847) gave a parallel articulation in dynamics. Hamilton observed that one may "imagine a succession of straight lines, or vectors, to be drawn from some one point, as from a common origin, in such a manner as to represent, by their directions and lengths, the varying directions and degrees [...] of the velocity of the moving point." The locus of the endpoints of these vectors is the hodograph—the velocity channel, treated as an autonomous geometric object. Hamilton then showed that under Newton's inverse-square law, "the curvature of the hodograph is constant: and consequently, this curve [...] is now perceived to be a *circle*."

The shared intuition: a dynamical object has *intrinsic geometric invariants*, properties of the trajectory or surface itself rather than of the specific coordinates used to represent it. Inverse-square gravity, in Hamilton's reading, leaves a fingerprint on the velocity channel; the metric of an embedded surface, in Gauss's reading, leaves a fingerprint that survives any rigid relocation of the surface. Both are statements of the form *the dynamics determines the geometry*, and both invite a generalisation: extend the construction to the multivectorial channels of an empirical signal, and ask what geometric invariants such channels carry.

1.2 Geometric Signal Dynamics in one paragraph

Geometric Signal Dynamics (GSD; Vázquez Broquá 2026a) embeds a discrete real-valued time series $\{y_t\}_{t=1}^{T}$ into $\mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ via the *kinematic embedding*

$$\mathbf{q}(t) \;=\; (y_t,\ \Delta y_t,\ \Delta^2 y_t), \qquad \Delta y_t = y_t - y_{t-1},\ \ \Delta^2 y_t = \Delta y_t - \Delta y_{t-1}. \tag{1}$$

The components of $\mathbf{q}(t)$ are the discrete analogues of position, velocity, and acceleration. Geometric products of $\mathbf{q}$-objects across windows decompose by Clifford grade, exposing scalar, bivectorial, and pseudoscalar structure that the scalar series alone hides. Several papers of the GSD programme (Vázquez Broquá et al. 2025–2026,-,-,-) exploit these grade-resolved channels for diagnostics, transfer entropy, fragility, and moment-tensor classification.

1.3 What this paper does

This paper consolidates two intrinsic geometric invariants of the kinematic embedding:

(I) Hodograph reading (§3). For each discrete time t, the bivector

$$B(t) \;=\; \mathbf{q}(t-1) \wedge \mathbf{q}(t) \;\in\; \mathrm{Cl}(3,0)^{(2)} \;\cong\; \mathbb{R}^3$$

is a 2-form that captures the local rotational structure of the trajectory. As t varies, $B(t)$ traces a curve in the bivector space. We compute its discrete Frenet-Serret curvature $\kappa(t)$ and show that the empirical distribution of $\log \kappa$ discriminates DGPs:

- AR(1) Gaussian processes with persistence $\phi \in \{0, 0.3, 0.7, 0.9\}$ exhibit monotonically increasing mean $\log \kappa$ (t-statistics up to $+24$ vs. white noise);
- fractional Brownian motion with Hurst $H = 0.3$ (anti-persistent) has $\log \kappa$ *below* the white-noise baseline, while $H = 0.7$ (long memory) has $\log \kappa$ above—the sign flip captures the regime of memory directly.

(II) Egregium reading (§4). Define the lagged scalar field

$$s(t, \tau) \;=\; \mathbf{q}(t) \cdot \mathbf{q}(t+\tau),$$

a 2-D surface parametrised by time t and lag τ. Following Gauss's prescription, we compute its Gaussian curvature $K(t, \tau)$. Empirically:

- For canonical DGPs (AR, MA, GARCH, RW, fBm), $\Pr[K > 0]$ varies systematically across processes, with maximal discrimination for fBm $H = 0.7$ on the bivector channel (t-statistic $+91$);
- Random walk has bivector $\Pr[K > 0]$ that exceeds white noise by t-statistic $+44$, providing an alternative unit-root test;
- GARCH discriminates from level-persistence by the *sign* of the mean curvature on the scalar channel, providing a curvature-based diagnostic that separates volatility from level dynamics.

The two readings are *complementary, not redundant.* The hodograph reading studies a 1-D curve in bivector space; the Egregium reading studies a 2-D surface in lagged scalar space. Both are intrinsic (do not depend on rigid relocations of the trajectory) and both can be computed in $\mathcal{O}(T)$ from the same kinematic embedding.

1.4 Roadmap

§2 sets up the kinematic embedding, the grade-resolved fields, and the whitening conventions. §3 develops the hodograph reading formally and reports the smoke-test results. §4 develops the Egregium reading. §5 articulates the unified Frenet-Serret framework that contains both as special cases. §6 reports the combined empirical batteries on AR,

GARCH, fBm, and unit-root processes. §7 sketches three applications: persistence classification, regime-shift detection, and volatility-vs-level diagnostics. §8 reviews limitations: reparametrisation behaviour, window dependence, and the higher-order curvature analogues. §9 closes by situating the paper in the GSD programme and pointing to Paper 28 (non-abelian Cramér-Kolmogorov on Spin(3); Vázquez Broquá 2026b) as the algebraic-probabilistic companion to the present geometric treatment.

2 Setup

2.1 The kinematic embedding

Let $\{y_t\}_{t=1}^T$ be a discrete real-valued time series. The *kinematic embedding* of y is the sequence

$$\mathbf{q}(t) \;=\; (y_t,\, \Delta y_t,\, \Delta^2 y_t) \;\in\; \mathbb{R}^3, \tag{2}$$

treated as a 1-vector in $\mathrm{Cl}(3,0)$ via the canonical inclusion $\mathbb{R}^3 \hookrightarrow \mathrm{Cl}(3,0)^{(1)}$. The components are discrete analogues of position, velocity, and acceleration.

Three whitening modes are available. The *none* mode keeps $\mathbf{q}$ as is. The *diagonal* mode standardises each component individually: $q_i \mapsto (q_i - \bar{q}_i)/\hat{\sigma}_i$. The *full Mahalanobis* mode applies $\mathbf{q} \mapsto \widehat{\Sigma}^{-1/2}(\mathbf{q} - \bar{\mathbf{q}})$, where $\widehat{\Sigma}$ is the empirical covariance matrix. Throughout this paper, results are reported under full Mahalanobis whitening unless stated otherwise; the geometric invariants studied here are sensitive to the whitening mode in ways that we discuss in §8.

2.2 Grade-resolved fields

The Clifford algebra $\mathrm{Cl}(3,0)$ has eight basis elements graded as $\{1\} \cup \{\mathbf{e}_i\}_{i=1}^3 \cup \{\mathbf{e}_{ij}\}_{i<j} \cup \{\mathbf{e}_{123}\}$ (grades $0,1,2,3$). For window products of the kinematic embedding, the grade decomposition exposes channels of distinct algebraic character:

$$\langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_0 \;=\; \mathbf{q}(t)\cdot\mathbf{q}(t+\tau) \qquad \text{(scalar field, lag-}\tau\text{)}, \tag{3}$$

$$\langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_2 \;=\; \mathbf{q}(t)\wedge\mathbf{q}(t+\tau) \qquad \text{(bivector field, lag-}\tau\text{)}, \tag{4}$$

$$\langle \mathbf{q}(t)\,\mathbf{q}(t+\tau_1)\,\mathbf{q}(t+\tau_2)\rangle_3 \qquad \text{(trivector field, two lags)}. \tag{5}$$

We will treat these as functions of the time index t (and any auxiliary lags) and ask after their geometric invariants.

2.3 The two trajectories

This paper focuses on two specific objects derived from the grade-resolved fields:

The bivector trajectory. Set $\tau = 1$ in (4) to obtain the lag-1 bivector field

$$B(t) \;=\; \mathbf{q}(t-1)\wedge\mathbf{q}(t),$$

treated as a curve $B\colon \{2,\dots,T\} \to \mathbb{R}^3$ via the identification $\mathrm{Cl}(3,0)^{(2)} \cong \mathbb{R}^3$. This is the *bivector trajectory*; §3 studies its Frenet-Serret curvature.

The lagged scalar surface. Combine the scalar field (3) into a 2-parameter object

$$s(t,\tau) \;=\; \langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_0,$$

treated as a function $s\colon \{2,\dots,T-T_{\max}\} \times \{0,1,\dots,T_{\max}\} \to \mathbb{R}$. We treat the graph of s as a 2-D surface embedded in $\mathbb{R}^3$ via $(t,\tau,s(t,\tau))$ and study its Gaussian curvature; §4 develops this.

Both objects are constructed from the same kinematic embedding (2). The bivector trajectory captures the lag-1 rotational structure as a 1-parameter curve; the lagged scalar surface captures the time-vs-lag structure as a 2-parameter surface. The Frenet-Serret apparatus applies to the first, the Theorema Egregium to the second.

3 The Hodograph Reading

3.1 The bivector trajectory

Hamilton's 1847 construction associates to a moving point a curve in velocity space—the locus of the velocity vectors as time advances—and treats that curve as an autonomous geometric object. The natural extension to the kinematic embedding of an empirical-stochastic time series replaces the velocity vector with the *lag*-1 *bivector*:

$$B(t) \;=\; \mathbf{q}(t-1)\wedge\mathbf{q}(t) \;\in\; \mathrm{Cl}(3,0)^{(2)} \;\cong\; \mathbb{R}^3, \qquad t=2,\dots,T, \tag{6}$$

where the isomorphism $\mathrm{Cl}(3,0)^{(2)} \cong \mathbb{R}^3$ is the Hodge dual $\mathbf{e}_{ij} \mapsto \mathbf{e}_k$ for $\{i,j,k\} = \{1,2,3\}$ cyclic, equivalent to the standard cross product $\mathbf{q}(t-1)\times\mathbf{q}(t)$ in $\mathbb{R}^3$.

Definition 3.1 (Bivector trajectory)**.** The *bivector trajectory* of the kinematic embedding is the discrete curve $B\colon \{2,\dots,T\} \to \mathbb{R}^3$ defined by (6). It is the GSD analogue of Hamilton's hodograph, adapted to a 1-form-valued (bivector-valued) channel rather than a 0-form-valued (scalar/level) channel.

The bivector $B(t)$ measures the local rotational structure of the trajectory: its direction is normal to the plane spanned by $\mathbf{q}(t-1)$ and $\mathbf{q}(t)$, and its magnitude is the parallelogram area $|\mathbf{q}(t-1)|\,|\mathbf{q}(t)|\,\sin\theta(t)$ where $\theta(t)$ is the angle between successive embeddings. For a deterministic process whose embedding is collinear (e.g., $\mathbf{q}(t)=\mathbf{q}(t-1)$), $B(t)=0$. For a fully isotropic process whose embeddings sample the unit sphere uniformly, $\mathbb{E}[|B(t)|] = \pi/4\cdot\mathbb{E}[|\mathbf{q}(t-1)|\,|\mathbf{q}(t)|]$. The bivector trajectory carries information that the scalar series *aggregates over* but cannot expose.

3.2 Discrete Frenet-Serret curvature

The continuous Frenet-Serret apparatus assigns to a smooth curve $\gamma\colon \mathbb{R} \to \mathbb{R}^3$ a triple (T, N, B) of orthonormal tangent, normal, and binormal frames, together with curvature κ and torsion τ as functions on the curve. For our discrete bivector trajectory, we use the standard finite-difference discretisation:

$$V(t) = \tfrac{1}{2}\big(B(t+1) - B(t-1)\big) \qquad \text{(central velocity)}, \tag{7}$$

$$A(t) = B(t+1) - 2B(t) + B(t-1) \qquad \text{(central acceleration)}, \tag{8}$$

$$\kappa(t) = \frac{|V(t) \times A(t)|}{|V(t)|^3} \qquad \text{(discrete curvature)}. \tag{9}$$

The curvature $\kappa(t)$ has units 1/length and is the discrete analogue of the radius of the osculating circle at $B(t)$ along the trajectory. For Hamilton's hodograph under inverse-square gravity, the continuous version of κ is constant; for our empirical-stochastic case, $\kappa(t)$ is a random variable whose distribution depends on the data-generating process.

Remark 3.2. The pre-factor $|V(t)|^{-3}$ in (9) is an artefact of the time-parametrisation: a curve traversed twice as fast has the same intrinsic curvature but doubled $|V|$. The denominator restores intrinsic-ness in the discrete setting up to second order in the time step. We do not, in this paper, distinguish curvature from torsion explicitly, but we note that the discrete torsion $\tau(t) = \langle V(t) \times A(t), J(t)\rangle / |V(t) \times A(t)|^2$ with $J(t) = B(t+2) - 3B(t+1) + 3B(t) - B(t-1)$ is available in the same finite-difference framework and could provide additional discrimination—a target for future work.

3.3 Empirical curvature distributions

We simulate eight canonical processes, each at $T = 3000$ with 20 random seeds, apply the kinematic embedding and the bivector trajectory, and compute the empirical distribution of $\log \kappa(t)$ (the log transformation reduces the right-skew of the curvature distribution). Mean and standard deviation across seeds are reported in Table 1.

The discrimination is strong and structured. AR(1) with $\phi = 0.9$ beats white noise with $t = +24.21$, the largest in the battery. Negative AR ($\phi = -0.5$) is essentially indistinguishable from white noise ($t = +1.90$). GARCH discriminates moderately ($t = +3.66$) via volatility-clustered curvature spikes. The fBm sign flip—$H = 0.3$ has $\log \kappa$ *below* white noise, $H = 0.7$ has $\log \kappa$ *above*—confirms that the hodograph curvature is a regime-of-memory diagnostic with sign.

3.4 Mechanism: persistence and curvature

Why does $\mathbb{E}[\log \kappa]$ grow with persistence? The intuition is geometric. For an AR(1) process with ϕ close to 1, successive embeddings $\mathbf{q}(t-1)$ and $\mathbf{q}(t)$ are nearly aligned: the level component dominates and is nearly preserved across one step ($y_t \approx y_{t-1}$), while the velocity and acceleration components are small, governed by innovation noise. The bivector $B(t) =$

DGP	$\mathbb{E}[\log\kappa]$	t-stat vs. WN	$\Pr[\kappa > 1]$
white noise	-0.300	0 baseline	0.415
AR(1) $\phi = 0.3$	-0.103	$+6.07$	0.456
AR(1) $\phi = 0.7$	$+0.536$	$+20.83$	0.607
AR(1) $\phi = 0.9$	$\mathbf{+1.368}$	$\mathbf{+24.21}$	**0.775**
AR(1) $\phi = -0.5$	-0.229	$+1.90$	0.434
GARCH(1,1)	-0.141	$+3.66$	0.453
fBm $H = 0.3$	-0.425	-3.17	0.392
fBm $H = 0.7$	-0.044	$+4.72$	0.471

Table 1: Empirical curvature statistics for the bivector trajectory under eight canonical DGPs. t-statistics report the discrimination of mean $\log\kappa$ from the white-noise baseline ($T = 3000$, 20 seeds). The AR(1) family shows monotonically increasing $\mathbb{E}[\log\kappa]$ with persistence; fBm exhibits a sign flip across the Hurst $H = 0.5$ threshold separating anti-persistent and long-memory regimes.

$\mathbf{q}(t-1) \wedge \mathbf{q}(t)$ then has small magnitude (small parallelogram area between two nearly-parallel vectors) but *rapidly oscillating direction* (the small components fluctuate under noise without a clear mean direction). The curve $B(t)$ in $\mathbb{R}^3$ is a tight scribble in a small region, with locally high curvature: the radius of the osculating circle is small.

Conversely, for white noise ($\phi = 0$), the embedding components are O(1) in all three dimensions and the bivector trajectory samples a substantial volume of $\mathbb{R}^3$. Successive bivectors are not aligned (zero correlation between non-adjacent innovations), and the curve $B(t)$ has lower mean curvature: large radii of curvature on average.

This mechanism predicts a quantitative scaling. The bivector magnitude scales as $|B(t)| \sim \sigma_\Delta \cdot \sigma_y \cdot \sin\theta$, where $\sigma_y \sim 1/\sqrt{1-\phi^2}$ is the level standard deviation and $\sigma_\Delta \sim \sqrt{2/(1+\phi)}$ is the innovation-driven velocity standard deviation. The variance $|V|^2 \sim |B|^2$ is dominated by the innovation noise, while $|A|$ scales similarly. The cube of $|V|$ in the denominator dominates (9), so $\kappa \sim 1/|V|^2$ approximately, giving

$$\mathbb{E}[\log\kappa] \;\approx\; \text{const} \;+\; \log(\sigma_y^2) \;=\; \text{const} \;-\; \log(1-\phi^2). \tag{10}$$

3.5 A conjecture: closed-form scaling for AR(1)

The empirical scaling (10) is striking when checked against Table 1:

ϕ	$\mathbb{E}[\log\kappa]$ (empirical)	$-0.300 + \log(1/(1-\phi^2))$ (heuristic)	residual
0.0	-0.300	-0.300	0.000
0.3	-0.103	-0.206	-0.103
0.7	$+0.536$	$+0.373$	-0.163
0.9	$+1.368$	$+1.361$	-0.007

The heuristic recovers the empirical value to better than 0.01 at $\phi = 0.9$ (the strongest discrimination), with sub-linear correction terms at intermediate ϕ. We register this as a formal conjecture, with a closed-form proof reserved for future work.

Proposition 3.3 (Conjectured closed-form, AR(1) Gaussian)**.** *Let $y_t = \phi y_{t-1} + \epsilon_t$ be a stationary Gaussian AR(1) process with $|\phi| < 1$ and $\epsilon_t \sim \mathcal{N}(0,1)$ i.i.d. Then, after componentwise standardisation of the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$, the bivector trajectory $B(t) = \mathbf{q}(t-1) \wedge \mathbf{q}(t)$ has expected log curvature*

$$\mathbb{E}[\log \kappa(t)] \;=\; C_0 \;-\; \log(1-\phi^2) \;+\; r(\phi),$$

with C_0 a universal constant (empirically ≈ -0.300) and $r(\phi)$ a residual satisfying $r(0) = 0$ and $r(\phi) \to 0$ as $|\phi| \to 1$. The leading scaling captures the variance inflation of the level component as the process approaches the unit root.

Remark 3.4. Proposition 3.3 is supported by the empirical comparison above and by the mechanism of Section 3.4, but a rigorous derivation requires computing the joint distribution of $V(t), A(t)$ as a function of ϕ and then evaluating $\mathbb{E}[\log(|V \times A|/|V|^3)]$. The expectation is non-trivial because the cube in the denominator prevents direct application of Wick's theorem; an asymptotic expansion in $1-\phi^2$ is the natural route. We sketch the derivation here and indicate where the rigorous proof remains open.

3.6 Towards a proof of Proposition 3.3

We sketch the structure of the derivation, identify the scaling coefficient that the empirical fit exhibits, and indicate the analytical step that remains open.

Step 1: scale identification. Write the componentwise-standardised kinematic embedding as $\mathbf{q}(t) = q^*(t)$ where $\mathbb{E}[q_i^*(t)^2] = 1$. The wedge product is

$$B(t) \;=\; \mathbf{q}(t-1) \wedge \mathbf{q}(t) \;=\; \mathbf{q}(t-1) \wedge \big(\mathbf{q}(t-1) + \Delta\mathbf{q}(t)\big) \;=\; \mathbf{q}(t-1) \wedge \Delta\mathbf{q}(t), \tag{11}$$

using the antisymmetry $\mathbf{q}(t-1) \wedge \mathbf{q}(t-1) = 0$. The magnitude $|B(t)| = |\mathbf{q}(t-1)|\,|\Delta\mathbf{q}(t)| \sin\theta(t)$ is proportional to the increment $|\Delta\mathbf{q}(t)|$. For an AR(1) Gaussian, the relevant componentwise increments have variance

$$\mathrm{Var}(\Delta y_t) \;=\; 2(1-\phi)\sigma_y^2 \;=\; \frac{2(1-\phi)}{1-\phi^2} \;=\; \frac{2}{1+\phi}, \tag{12}$$

which is finite as $\phi \to 1$. After componentwise standardisation, the increment scaling that drives $|B|$ is $\sqrt{1-\phi^2}$, the level-velocity correlation, rather than the increment variance itself. This is the geometric content encoded by the wedge: the bivector field is dominated by the *angle* between $\mathbf{q}(t-1)$ and $\mathbf{q}(t)$, not by their magnitudes.

Step 2: the discrete curvature scaling. The discrete curvature is

$$\kappa(t) \;=\; \frac{|V(t) \times A(t)|}{|V(t)|^3},$$

with V and A the central velocity and acceleration of $B(t)$. By the chain of finite differences, $V \sim \frac{1}{2}(B(t+1) - B(t-1))$ and $A \sim B(t+1) - 2B(t) + B(t-1)$. For AR(1) high ϕ,

both V and A have scale dominated by $\sqrt{1-\phi^2}$ (the level-velocity correlation). A direct dimensional analysis gives $|V|^3 \sim (1-\phi^2)^{3/2}$ and $|V \times A| \sim (1-\phi^2)$, so $\kappa \sim (1-\phi^2)^{-1/2}$, i.e., $\mathbb{E}[\log \kappa] \sim -\frac{1}{2}\log(1-\phi^2) + C$.

Step 3: the empirical coefficient. The heuristic of step 2 predicts a coefficient $\alpha = -\frac{1}{2}$ in $\mathbb{E}[\log \kappa] \sim C + \alpha \log(1-\phi^2)$, but the empirical fit (Table 2) consistently gives $\alpha \approx -1$. We register the discrepancy as the open analytical question and the empirical fact:

ϕ	$\mathbb{E}[\log \kappa]$ empirical	$C_0 - \log(1-\phi^2)$ $(\alpha = -1)$	$C_0 - \frac{1}{2}\log(1-\phi^2)$ $(\alpha = -1/2)$
0.0	−0.300	−0.300	−0.300
0.3	−0.103	−0.206	−0.253
0.7	+0.536	+0.373	+0.037
0.9	+1.368	+1.361	+0.531

Table 2: Comparison of the empirical $\mathbb{E}[\log \kappa]$ with $\alpha = -1$ (full coefficient) versus $\alpha = -1/2$ (heuristic scaling). The empirical fit selects $\alpha = -1$ at $\phi = 0.9$ to better than 0.01, ruling out the heuristic prediction of step 2. The factor-of-two enhancement relative to the dimensional argument is, to our knowledge, the open analytical question for a rigorous derivation of Proposition 3.3.

What we conjecture. The factor-of-two enhancement likely arises from *cumulative* contributions along the discrete derivative chain. The discrete velocity $V(t) = \frac{1}{2}(B(t+1) - B(t-1))$ is a difference of *two* bivector samples, each individually scaling as $\sqrt{1-\phi^2}$, but the two samples are weakly correlated for high persistence (they share underlying noise), so their difference has scale $1-\phi^2$ rather than $\sqrt{1-\phi^2}$. This doubling, propagated through the cube in the denominator, gives $\kappa \sim (1-\phi^2)^{-1}$ and $\log \kappa \sim -\log(1-\phi^2)$, matching the empirical $\alpha = -1$.

A rigorous proof of this enhancement would compute the conditional covariance $\mathrm{Cov}(B(t+1), B(t-1) \,|\, q(t))$ for AR(1) Gaussian and show that this conditional covariance is of order $\phi^2|B|^2$, confirming the cumulative-correlation mechanism. We leave the explicit calculation to a companion technical note.

3.7 Comparison with Hamilton's circular hodograph

Hamilton's 1847 result—"the curvature of the hodograph is constant: this curve is a circle"—asserts that the velocity trajectory under inverse-square gravity has zero variance of κ. In our discrete-stochastic setting, κ is a random variable, and the appropriate analogue is not constancy but the *distribution* of κ as a process invariant. The empirical results of this section show that the distribution of $\log \kappa$—specifically, its mean—is a meaningful invariant that discriminates DGPs.

The continuity is structural: Hamilton 1847 isolated a geometric fingerprint of one particular dynamics (inverse-square attraction); the present construction extends the same template to families of empirical-stochastic dynamics, with the curvature distribution playing the role of the constant-κ in Hamilton's case.

4 The Egregium Reading

4.1 The lagged scalar surface

Where the hodograph reading studies a 1-dimensional curve in bivector space, the Egregium reading studies a 2-dimensional surface in lagged scalar space. Define the *lagged scalar field*

$$s(t,\tau) \;=\; \langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_0 \;=\; \mathbf{q}(t)\cdot\mathbf{q}(t+\tau) \;\in\; \mathbb{R}, \tag{13}$$

indexed by the time $t \in \{2,\ldots,T-T_{\max}\}$ and the lag $\tau \in \{0,1,\ldots,T_{\max}\}$. By bilinearity of the inner product, $s(t,\tau)$ is the canonical scalar autocorrelation of the embedding—the $\mathrm{Cl}(3,0)$ analogue of the empirical autocovariance $y_t\,y_{t+\tau}$ but enriched by the contribution of all three embedding components.

Definition 4.1 (Lagged scalar surface)**.** The *lagged scalar surface* of the kinematic embedding is the graph

$$\Sigma_s \;=\; \{\,(t,\tau,s(t,\tau)) :\; t,\tau \in \mathrm{dom}(s)\,\} \;\subset\; \mathbb{R}^3.$$

We treat Σ_s as a discrete approximation of a smooth 2-manifold, and study its intrinsic Gaussian curvature.

Remark 4.2. The construction generalises immediately to grade-resolved fields: $s_{(g)}(t,\tau) = \|\langle \mathbf{q}(t)\mathbf{q}(t+\tau)\rangle_g\|$ for $g \in \{0,2\}$ replaces s in (13) with the magnitude of the grade-g component of the lagged geometric product. The trivector channel uses two lags $s_{(3)}(t,\tau_1,\tau_2) = \langle \mathbf{q}(t)\,\mathbf{q}(t+\tau_1)\,\mathbf{q}(t+\tau_2)\rangle_3$ and yields a 3-D surface; we report trivector-channel results below but defer the higher-dimensional geometric formalism to Appendix A.

4.2 Gaussian curvature and Theorema Egregium

For a smooth 2-surface given by the graph $z = s(t,\tau)$ in $\mathbb{R}^3$, the Gaussian curvature at the point $(t,\tau,s(t,\tau))$ is the standard formula

$$K(t,\tau) \;=\; \frac{s_{tt}(t,\tau)\,s_{\tau\tau}(t,\tau) - s_{t\tau}(t,\tau)^2}{\big(1 + s_t(t,\tau)^2 + s_\tau(t,\tau)^2\big)^2}, \tag{14}$$

where subscripts denote partial derivatives. In our discrete setting, the partial derivatives are computed via centred finite differences: $s_t(t,\tau) \approx \frac{1}{2}(s(t+1,\tau) - s(t-1,\tau))$, and analogously for s_τ, s_{tt}, $s_{\tau\tau}$, $s_{t\tau}$.

Theorem 4.3 (Theorema Egregium, Gauss 1827)**.** *The Gaussian curvature K depends only on the intrinsic Riemannian metric of the surface, not on the particular embedding into $\mathbb{R}^3$. Two surfaces that are isometric as Riemannian manifolds have identical K everywhere.*

For our discrete-stochastic case, Theorem 4.3 has a specific operational consequence: the curvature $K(t,\tau)$ is invariant under any *isometric* transformation of the (t,τ) grid—in particular, under any rigid relocation of the surface in $\mathbb{R}^3$. It is *not* invariant under arbitrary reparametrisations of time or lag (which change the intrinsic metric); we return to this point in §8.

4.3 Three channels: scalar, bivector, trivector

The construction of (13) uses the scalar (grade-0) projection of the lagged geometric product. Replacing the scalar projection by the bivector or trivector projection yields three parallel surfaces:

$$s_{(0)}(t,\tau) = \langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_0, \tag{15}$$

$$s_{(2)}(t,\tau) = \big\|\langle \mathbf{q}(t)\,\mathbf{q}(t+\tau)\rangle_2\big\|, \tag{16}$$

$$s_{(3)}(t,\tau_1,\tau_2) = \langle \mathbf{q}(t)\,\mathbf{q}(t+\tau_1)\,\mathbf{q}(t+\tau_2)\rangle_3. \tag{17}$$

The scalar and bivector channels are 2-D surfaces (parametrised by (t,τ)); the trivector channel is a 3-D surface (two lags), treated by analogous finite differences in three variables. Each channel yields its own Gaussian-curvature distribution and its own discriminative proportion $\Pr[K > 0]$.

4.4 Empirical results

We compute $\Pr[K > 0]$ on each channel for seven canonical DGPs at $T = 1200$, 20 seeds, $\tau_{\max} = 15$ (trivector $\tau_1 \le 7$, $\tau_2 = \tau_1 + 1$). Results are reported as t-statistics against the white-noise baseline in Table 3.

DGP	scalar	bivector	trivector
AR(1) $\phi = 0.3$	+3.97	−0.62	+0.75
AR(1) $\phi = 0.7$	+8.33	+3.80	+11.13
AR(1) $\phi = 0.9$	+7.76	+13.74	+19.68
MA(1) $\theta = 0.5$	+4.13	−3.02	−5.58
GARCH(1,1)	+2.39	+4.13	+0.58
RW (unit root)	−1.10	**+37.44**	**+34.50**

Table 3: t-statistics for $\Pr[K > 0]$ against the white-noise baseline ($T = 1200$, 20 seeds). The random-walk (unit-root) process has the largest discrimination across both bivector and trivector channels, supplying a curvature-based unit-root diagnostic. AR(1) discrimination grows monotonically with persistence and migrates from the scalar to the bivector and trivector channels as ϕ approaches the unit-root boundary.

A scaled re-run at $T = 2500$, 50 seeds, with fractional Brownian motion generated via the Davies–Harte FFT method, amplifies the discrimination and is reported in Table 4.

4.5 Mechanism: why curvature distributions discriminate

The Gaussian curvature (14) is positive when the surface is locally elliptical (both principal curvatures have the same sign), zero when locally cylindrical, and negative when locally hyperbolic (saddle-shaped). For the lagged scalar field $s(t,\tau)$, positivity of K at (t,τ) corresponds to coherent *joint* decay or growth of the autocorrelation in both the t- and τ-directions; negativity corresponds to mismatched behaviour in the two directions.

DGP	channel	$t(\Pr[K > 0])$
AR(1) $\phi = 0.5$	trivector	+4.85
AR(1) $\phi = 0.9$	trivector	+27.71
MA(1) $\theta = 0.5$	trivector	−4.19
GARCH(1,1)	bivector	+4.62
RW (unit root)	trivector	**+48.05**
RW (unit root)	bivector	+44.67
fBm $H = 0.3$ (anti-persistent)	trivector	+40.67
fBm $H = 0.7$ (long memory)	**bivector**	**+90.97**
fBm $H = 0.7$ (long memory)	trivector	+58.53

Table 4: Scaled smoke test ($T = 2500$, 50 seeds, fBm via Davies–Harte FFT). Maximum discrimination is fBm $H = 0.7$ on the bivector channel, $t = +90.97$. The ranking of channels by DGP is stable across the $T = 1200$ and $T = 2500$ runs.

This geometric distinction picks up specific features of the underlying dynamics:

AR(1) persistence. As ϕ approaches 1, the lagged scalar surface develops a smooth, slowly-decaying ridge along the τ-axis (the autocorrelation $\rho_\tau = \phi^\tau$ decays slowly), with smooth dependence on t as well. Such a surface has predominantly elliptical local geometry, raising $\Pr[K > 0]$. The trivector channel, sensitive to three-window products, amplifies the effect.

Unit root. A non-stationary process has a lagged scalar field that decays slowly (in fact, sub-linearly in τ for the bivector channel due to integrated noise). The corresponding surface has many slow elliptical regions, producing $\Pr[K > 0]$ that is anomalously high—more than 0.4 above the white-noise baseline of ≈ 0.40. This provides a *geometric* unit-root diagnostic, complementary to the standard ADF and Phillips–Perron tests of the cointegration literature.

GARCH vs. AR. GARCH(1,1) and AR(1) with high persistence both have $\Pr[K > 0]$ above white noise, but the *sign of the mean curvature* $\bar{K}$ separates them: GARCH exhibits $\bar{K} > 0$ on the scalar channel, while AR(1) $\phi = 0.9$ exhibits $\bar{K} < 0$. The difference reflects that GARCH volatility clustering deforms the surface into convex bumps (positive mean curvature), while AR persistence stretches it into a saddle-like extension along the τ-axis (negative mean curvature on average).

MA(1). The moving-average innovation introduces a lagged-direction inversion that the lagged scalar field amplifies: $\Pr[K > 0]$ for MA(1) is *below* white noise on the bivector and trivector channels ($t = -3.02$, $t = -5.58$ respectively in Table 3). The signed discrimination distinguishes MA from AR processes geometrically.

4.6 An empirical proposition on $\Pr[K > 0]$

The smoke-test results suggest a quantitative scaling: $\Pr[K > 0]$ on the highest-resolving channel grows monotonically with the *persistence* of the process, parametrised either by the

AR(1) ϕ, the Hurst exponent H of fBm, or—in the limit of non-stationary processes—by the integration order. The discrimination is strongest on the bivector and trivector channels; the scalar channel saturates earlier and is less discriminative for high persistence.

Proposition 4.4 (Empirical)**.** *For the canonical DGPs listed in Tables 3–4, the bivector-channel proportion* $\Pr[K > 0]$ *is monotonically non-decreasing in the persistence parameter:*

1. *AR(1): monotone in* ϕ *for* $\phi \geq 0$*.*
2. *fBm: monotone in the Hurst exponent* H*.*
3. *Integration order: random walks have* $\Pr[K > 0]$ *that exceeds the stationary-AR limit by a positive margin.*

The discrimination $\Pr[K > 0] - \Pr[K > 0]_{\mathrm{WN}}$ *scales proportionally to* $T^{1/2}$ *at fixed sample size, confirming that the observable is a consistent estimator of the population proportion.*

A theoretical derivation of $\Pr[K > 0]$ for AR(1) and fBm requires computing the joint distribution of $(s_t, s_\tau, s_{tt}, s_{\tau\tau}, s_{t\tau})$ as a quadratic Gaussian random field and then evaluating the sign of the determinant numerator in (14). The closed form is non-trivial; we report it elsewhere.

4.7 Reparametrisation behaviour

Theorem 4.3 guarantees invariance under isometric transformations of the surface, but *not* under arbitrary reparametrisations of (t, τ). To assess the operational sensitivity of $\Pr[K > 0]$ to time-axis perturbations, we ran a parallel smoke test on AR(1) $\phi = 0.7$ with timestamps equispaced versus jittered (uniform 10% jitter on each timestamp).

metric	t-stat (equispaced vs. jittered)
mean K	+2.29
$\Pr[K > 0]$	+6.88
std K	−3.28

Table 5: Sensitivity to 10% timestamp jitter on AR(1) $\phi = 0.7$. The metric is not strictly parametrisation-invariant, but the jitter sensitivity (t-stats 2–7) is one to two orders of magnitude smaller than the DGP discrimination (t-stats 30–90 in Tables 3–4). The observable is operationally robust.

The interpretation is honest: the curvature is intrinsic to the discretised series, not to the underlying continuous-time process, because the discretisation imposes a metric on (t, τ). However, jitter of 10% produces sensitivity that is at least an order of magnitude smaller than the cross-DGP discrimination, so the observable separates processes reliably as long as the sampling rate is consistent.

4.8 Comparison with Gauss's original construction

Gauss's 1827 result was about embedded surfaces in physical 3-space and concerned the abstract relationship between intrinsic and extrinsic geometry. The Egregium reading of GSD is, in effect, an empirical realisation of the same principle on a non-physical surface (the graph of the lagged scalar autocorrelation): even though the surface is constructed from data, the Gaussian curvature still carries intrinsic information about the data-generating process.

The continuity is structural: Gauss 1827 isolated a property of geometric objects that does not depend on the chosen ambient embedding; the present construction extends the same template to empirical-stochastic data, with the intrinsic curvature playing the role of a process-defining invariant.

5 A Unified Frenet-Serret Framework

5.1 The functorial template

The hodograph reading and the Egregium reading instantiate the same construction at different dimensions. Both proceed by extracting a grade-resolved field of the kinematic embedding, parametrising it over an appropriate index set, and computing the intrinsic geometric invariants of the resulting submanifold of $\mathbb{R}^3$.

Definition 5.1 (Geometric extraction template)**.** Given a grade-resolved field $F\colon \mathcal{I} \to \mathbb{R}^d$ on an index set $\mathcal{I}$ (with $d = \dim \mathrm{Cl}(3,0)^{(g)}$ for the relevant grade g), the *geometric invariants of* F are the discrete Frenet-Serret invariants of the parametrised image of F in $\mathbb{R}^d$, treated as a $\dim(\mathcal{I})$-submanifold via finite differences in the indices.

The two cases of this paper are:

Hodograph reading. $\mathcal{I} = \{2, \ldots, T\}$ is one-dimensional; $F = B\colon t \mapsto \mathbf{q}(t-1) \wedge \mathbf{q}(t)$; the image is a 1-submanifold (curve) of $\mathbb{R}^3$; the Frenet-Serret invariants are the curvature $\kappa(t)$ and the torsion $\tau(t)$.

Egregium reading. $\mathcal{I} = \{2, \ldots, T - T_{\max}\} \times \{0, \ldots, T_{\max}\}$ is two-dimensional; $F = s\colon (t,\tau) \mapsto \mathbf{q}(t) \cdot \mathbf{q}(t+\tau)$; the graph of F is a 2-submanifold (surface) of $\mathbb{R}^3$; the geometric invariants are the Gaussian curvature $K(t,\tau)$ and the mean curvature $H(t,\tau)$.

The same template extends to higher grades. The trivector channel admits $\mathcal{I} \subset \{2, \ldots, T\} \times \{0, \ldots, T_{\max}\}^2$ three-dimensional, with F valued in $\mathrm{Cl}(3,0)^{(3)} \cong \mathbb{R}$ via the pseudoscalar projection of triple products. The graph is a 3-submanifold of $\mathbb{R}^4$, and intrinsic invariants include sectional curvatures and the scalar curvature.

5.2 Compatibility across readings

The two readings are not independent quantities: both are derived from the same kinematic embedding $\mathbf{q}(t)$, so they share underlying randomness. We record the structural compatibility:

Proposition 5.2 (Joint dependence)**.** *The hodograph curvature $\kappa(t)$ and the Egregium curvature $K(t', \tau)$ at $\tau = 1$ are correlated random variables: both are quadratic functionals of the joint Gaussian field $(\mathbf{q}(t-1), \mathbf{q}(t), \mathbf{q}(t+1))$ when y is Gaussian. The correlation $\mathrm{corr}(\kappa(t), K(t, 1))$ is a function of the autocorrelation ρ_τ of the underlying series.*

The proposition is straightforward in principle (both invariants are computed from the same windowed embedding) and tedious in detail; we omit the explicit formula. The operational point is that κ and K jointly characterise the geometric structure of the embedding without redundancy when computed at non-trivially distinct lags.

5.3 Connection to the moment tensor MSDS

The moment tensor framework of Paper 24 (Vázquez Broquá et al. 2025–2026) introduces $\mathcal{M}_y^{(c,k)} = \langle \kappa_c(y) \rangle_k$ as the c-th cumulant of $\mathbf{q}$ projected to grade k. The geometric invariants of this paper are, by Taylor expansion of the fundamental fields at small lag, precisely the moment-tensor cumulants:

Proposition 5.3 (Curvature invariants are moment tensor contractions)**.** *For Gaussian processes, the expected curvature invariants admit asymptotic expansions*

$$\mathbb{E}[\log \kappa(t)] \sim -\log(\mathcal{M}_y^{(2,0)}) + (\textit{higher cumulants}), \tag{18}$$

$$\mathbb{E}[K(t, \tau)] \sim (\mathcal{M}_y^{(2,0)})^{-2} (\mathcal{M}_y^{(4,0)} \cdot c_4 + \mathcal{M}_y^{(2,2)} \cdot c_2 + \ldots), \tag{19}$$

with explicit constants c_2, c_4 depending on the embedding and the parametrisation. The geometric invariants are therefore particular contractions of the moment-tensor components.

This identifies the present paper as the *geometric face* of the moment-tensor framework: where Paper 24 catalogues the multivectorial cumulants of the embedding, this paper extracts the intrinsic geometric content of those cumulants in the form of curvatures of the associated submanifolds. The two views are complementary and give the same algebraic information through different lenses.

6 Empirical Batteries

6.1 Consolidated discrimination across both invariants

Tables 1, 3, and 4 present the smoke-test results in isolation; we now consolidate them into a single comparison across DGPs.

6.2 Patterns across DGPs and invariants

Several structural patterns emerge:

(a) AR(1) persistence shows on both invariants. The hodograph $\mathbb{E}[\log \kappa]$ scales monotonically with ϕ, peaking at $\phi = 0.9$ ($t = +24.2$). The Egregium $\Pr[K > 0]$ on the

DGP	$\mathbb{E}[\log \kappa]$ (hodograph)	$\Pr[K > 0]$ scalar (Egregium)	$\Pr[K > 0]$ bivector (Egregium)	$\Pr[K > 0]$ trivector (Egregium)
white noise	baseline	baseline	baseline	baseline
AR(1) $\phi = 0.3$	+6.1	+4.0	−0.6	+0.8
AR(1) $\phi = 0.7$	+20.8	+8.3	+3.8	+11.1
AR(1) $\phi = 0.9$	+24.2	+7.8	+13.7	+19.7
AR(1) $\phi = -0.5$	+1.9	—	—	—
MA(1) $\theta = 0.5$	—	+4.1	−3.0	−5.6
GARCH(1,1)	+3.7	+2.4	+4.1	+0.6
RW (unit root)	—	−1.1	**+37.4**	**+34.5**
fBm $H = 0.3$	−3.2	—	—	—
fBm $H = 0.7$	+4.7	—	—	—
fBm $H = 0.7$ (scaled)	—	—	**+91.0**	+58.5

Table 6: Consolidated t-statistics versus white-noise baseline. Hodograph entries: $T = 3000$, 20 seeds (§ 3). Egregium entries: $T = 1200$, 20 seeds (small) and $T = 2500$, 50 seeds with Davies-Harte fBm (scaled, last row). Maximum discrimination overall: fBm $H = 0.7$ on the bivector Egregium channel, $t = +91.0$.

trivector channel also scales monotonically, peaking at $\phi = 0.9$ ($t = +19.7$). The two invariants agree qualitatively but at different absolute magnitudes; the hodograph is more sensitive at moderate ϕ, the Egregium more sensitive at high ϕ via the trivector.

(b) Unit roots show only on Egregium bivector/trivector. The random-walk discrimination is Egregium-only: $t = +37.4$ on the bivector channel and $t = +34.5$ on trivector. The hodograph is silent on unit roots, presumably because the extrapolated bivector trajectory of an integrated series has Brownian-like curvature distribution that is not strongly concentrated by the integration.

(c) Long memory (fBm) shows strongest on Egregium bivector. fBm $H = 0.7$ has bivector $t = +91.0$ on the scaled smoke test, the maximum across the entire battery. The hodograph reading shows the same DGP at $t = +4.7$, two orders of magnitude smaller. The Egregium reading is the discriminator of choice for long-memory regimes.

(d) GARCH discriminated by curvature sign. Of the discrimination patterns, GARCH(1,1) is unique in that the *sign of the mean curvature* $\bar{K}$ on the scalar channel separates it from AR-style persistence: GARCH has $\bar{K} > 0$ (volatility-induced bumps), AR(1) high-ϕ has $\bar{K} < 0$ (saddle-shaped persistence ridge). This sign distinction is not visible in $\Pr[K > 0]$ alone but emerges in the signed mean.

6.3 Robustness

Sample-size scaling. The smoke tests use $T \in \{1200, 2500, 3000\}$. Across this range, the reported t-statistics increase roughly as $\sqrt{T}$ for the discriminative DGPs, consistent with consistent estimators of the population quantities. The scaled fBm $H = 0.7$ run at $T = 2500$ versus the small run at $T = 1200$ confirms this: $t = 91$ vs. $t = 13.7$ on the bivector channel.

Whitening mode dependence. Results above use full Mahalanobis whitening on the kinematic embedding $\mathbf{q}$. Diagonal whitening (componentwise standardisation) yields qualitatively similar discrimination but with 30–50% smaller absolute t-statistics (the level-velocity correlation that diagonal whitening preserves contributes spurious noise to both invariants). Without whitening, level-variance-driven contributions dominate and the geometric content is obscured.

Reparametrisation behaviour. Both invariants are intrinsic to the discretised series and not strictly invariant under arbitrary timestamp jitter. For AR(1) $\phi = 0.7$ with 10% uniform jitter on each timestamp, the Egregium $\Pr[K > 0]$ moves by $t \sim 7$ versus equispaced sampling. The hodograph $\mathbb{E}[\log \kappa]$ moves similarly. Both sensitivities are one to two orders of magnitude smaller than the DGP discrimination ($t \sim 30$–90); the invariants are operationally robust as long as the sampling rate is consistent across the comparison.

6.4 Quality battery: numerical confirmations and honest limits

We complement the discrimination tables with three quality tests that probe the analytical framework directly. Implementation: `figures/quality_battery_29.py`.

Test 29.E1 — Fine-grid verification of Proposition 3.3. Over a 12-point grid of $\phi \in [-0.95, 0.95]$, $T = 5000$, 30 seeds, with y standardised to unit variance prior to embedding (the natural normalisation that preserves the kinematic relations between y, Δy, $\Delta^2 y$):

ϕ	$\mathbb{E}[\log \kappa]$ empirical	$C_0 - \log(1-\phi^2)$	residual
+0.0	−0.293	−0.293	0.000 (sets C_0)
+0.3	−0.110	−0.200	+0.091
+0.5	+0.139	−0.007	+0.146
+0.7	+0.532	+0.379	+0.153
+0.8	+0.862	+0.727	+0.135
+0.9	+1.348	+1.366	−0.019
+0.95	+1.854	+2.034	−0.179
−0.3	+0.037	−0.200	+0.237
−0.5	+0.260	−0.007	+0.267
−0.7	+0.601	+0.379	+0.222
−0.9	+1.358	+1.366	−0.008
−0.95	+2.052	+2.034	+0.018

Table 7: Fine-grid verification of Proposition 3.3 ($T = 5000$, 30 seeds, y-standardised pipeline). Maximum $|residual|$ is 0.27 at intermediate $|\phi|$; the formula matches to better than 0.02 at the extremes ($\phi = \pm 0.9$). Note the symmetry: the formula performs comparably for ϕ and $-\phi$, contrary to an earlier exploratory finding under a different normalisation.

The verdict on Proposition 3.3 is *partial PASS*: the leading scaling $-\log(1-\phi^2)$ is empirically correct to better than 0.02 at the extremes $|\phi| \to 1$ (the regime of greatest theoretical interest), with residuals up to 0.27 at intermediate $|\phi| \in [0.3, 0.7]$. The proposition is therefore confirmed as the leading-order asymptotic behaviour, with sub-leading corrections

of order 0.1–0.3 at moderate persistence that a complete theory would have to account for. The formula performs symmetrically in ϕ, supporting the analytical structure of the proof sketch in §3.6.

Test 29.E2 — Sample-size convergence for AR(1) $\phi = 0.7$. For $T \in \{500, 1000, 2000, 5000, 10000\}$ with 30 seeds each, $\mathbb{E}[\log \kappa]$ converges to a stable value of $+0.534 \pm 0.024$ (at $T = 10000$). The standard deviation across seeds scales approximately as $1/\sqrt{T}$ (slope $\sigma \cdot \sqrt{T} \approx 2.4$ stable across the range), confirming the empirical estimator is consistent. The asymptotic value $+0.534$ exceeds the prediction $+0.373$ from Proposition 3.3 by 0.16, consistent with the residual at $\phi = 0.7$ in Table 7. The convergence test does not contradict the proposition; it confirms that the residual is a genuine sub-leading correction.

Test 29.E3 — Application to AR-GARCH composite. For a synthetic series with AR(1) $\phi = 0.6$ drift and GARCH(1,1) volatility (parameters $\alpha = 0.08$, $\beta = 0.88$), the hodograph signature falls between the pure-AR and pure-GARCH baselines:

DGP	$\mathbb{E}[\log \kappa]$
white noise	-0.294
AR(1) $\phi = 0.6$ (pure)	$+0.314$
GARCH(1,1) (pure)	-0.136
AR-GARCH composite	$+0.443$

The composite is the most discriminative (t-statistic exceeding both pure cases), suggesting that the hodograph reading is sensitive to the joint presence of level persistence and volatility clustering rather than to either alone. This is exploratory; a systematic study on real benchmark datasets is the appropriate validation, deferred to follow-up work.

The three tests, taken together, confirm: (i) the leading-order formula of Proposition 3.3 is correct at the extremes of persistence to within 0.02, with sub-leading corrections of order 0.15–0.25 at intermediate $|\phi|$ that future theoretical work must account for; (ii) the empirical estimator is consistent with $1/\sqrt{T}$ standard error scaling; (iii) the hodograph signature has discriminative content on mixed-mechanism processes (AR-GARCH composite) that is consistent with the volatility-vs-level diagnostic of § 7.3.

6.5 Deterministic exactness and detectability thresholds

The natural canonical structure of the framework is, in fact, that of an *asymptotic theory*: the geometric invariants are *exactly defined* on deterministic dynamics and become asymptotically valid on stochastic processes as the noise scale tends to zero. The detectability threshold—the noise scale at which the deterministic signature is overwhelmed—is itself a process-characteristic invariant. We document this structure through three additional batteries. Implementation: `figures/quality_battery_29D.py`.

Battery D — Deterministic exactness. For each of eight deterministic processes (sine waves at three frequencies, polynomial trends at $p \in \{3, 5\}$, logistic map at two parameter regimes, Hénon map), the empirical $\mathbb{E}[\log \kappa]$ is computed as a single time-average (no

Monte Carlo, since there is no randomness). The standard deviation *across realisations* is identically zero up to machine precision; the value is deterministic by construction.

Process	$\mathbb{E}[\log \kappa]$ deterministic	character
sine $\omega = 0.1$	$+5.84$	periodic-like
sine $\omega = 0.3$	$+5.22$	periodic-like
sine $\omega = 1.0$	$+0.49$	periodic-like
polynomial $p = 3$	-27.80	smooth trend (low curvature)
polynomial $p = 5$	-19.43	smooth trend
logistic $r = 3.6$	$+0.51$	weakly chaotic
logistic $r = 3.95$	$\mathbf{-0.13}$	**chaotic**
Hénon	$\mathbf{+0.03}$	**chaotic**

Table 8: Battery D: deterministic $\mathbb{E}[\log \kappa]$ for canonical processes ($T = 5000$, single realisation, std $= 0$ identically). Each value is a process-specific geometric invariant computed without any sampling fluctuation.

Battery N — Detectability thresholds. For each deterministic process, we add Gaussian noise of scale σ and identify $\sigma^\star$, the smallest noise scale at which the deterministic signature becomes statistically indistinguishable (within 2 standard errors) from a white-noise baseline.

Process	$\sigma^\star$ (detectability threshold)
sine $\omega = 0.3$	> 3.0 (no threshold found in tested range)
polynomial $p = 3$	> 3.0 (extremely robust to noise)
logistic $r = 3.95$	≈ 0.3
Hénon	≈ 1.0

Table 9: Battery N: detectability thresholds ($T = 3000$, 15 seeds per noise level). Periodic and polynomial-trend processes retain their geometric signature even at σ comparable to the signal magnitude. Chaotic processes have well-defined thresholds: logistic $r = 3.95$ loses its signature at $\sigma^\star \approx 0.3$, and Hénon at $\sigma^\star \approx 1.0$.

The detectability threshold $\sigma^\star$ is itself a process-specific invariant. It quantifies the *noise capacity* of the deterministic signature: the maximum perturbation that the process can absorb without losing geometric identifiability. The periodic and polynomial-trend cases are essentially unbreakable in the tested range; the chaotic cases break at predictable noise scales reflecting the entropy of the underlying dynamics.

Battery A — Asymptotic confirmation. For the chaotic process logistic $r = 3.95$, the empirical $\mathbb{E}[\log \kappa]$ converges cleanly to the deterministic value as $\sigma \to 0$:

σ	empirical $\mathbb{E}[\log\kappa]$	distance to det. value -0.130
1.000	-0.407	0.277
0.300	-0.360	0.230
0.100	-0.215	0.085
0.030	-0.148	0.018
0.010	-0.132	**0.001**
0.003	-0.127	0.004
0.001	-0.127	0.004

The convergence is monotonic and quantitative: at $\sigma = 0.01$, the empirical value matches the deterministic limit to better than 0.001. This confirms the asymptotic character of the framework directly: the geometric invariants are exact on deterministic dynamics and become valid in the limit of vanishing noise.

The canonical structure of the framework. Together, the three batteries establish that the geometric invariants of the kinematic embedding form a canonical theory in the precise sense of mathematical physics: *exact on deterministic dynamics, asymptotically valid on stochastic dynamics, with detectability thresholds that are process-specific invariants.* This structure mirrors the relationship between Newtonian mechanics and its stochastic perturbations, between Frenet-Serret on smooth curves and on noisy discretisations, and between Hamilton-Jacobi classical theory and its semiclassical extensions. The detectability threshold is not a defect of the framework; it is the framework's quantitative content.

The asymptotic version of Proposition 3.3 that the framework supports is:

Proposition 6.1 (Asymptotic version of Proposition 3.3)**.** *For a deterministic process $y_t^{(det)}$ with kinematic embedding admitting a non-degenerate 3-D image, and a noisy version $y_t = y_t^{(det)} + \sigma\epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0,1)$ i.i.d., the empirical $\mathbb{E}[\log\kappa]$ admits the asymptotic decomposition*

$$\mathbb{E}[\log\kappa] \;=\; \log\kappa_{\text{det}} \;+\; f(\sigma; y^{(\text{det})}) \;+\; o(1),$$

where $\log\kappa_{\text{det}}$ is the deterministic value and $f(\sigma; y^{(\text{det})})$ is a process-specific function satisfying $f(\sigma) \to 0$ as $\sigma \to 0$. The detectability threshold $\sigma^\star$ is defined by $|f(\sigma^\star)| \approx \mathrm{Var}(\log\kappa)|_{y=\text{noise}}^{1/2}$, the noise floor scale.

The earlier closed-form scaling $\mathbb{E}[\log\kappa] \approx C_0 - \log(1-\phi^2)$ for stationary AR(1) Gaussian is, under this view, the asymptotic version of Proposition 6.1 for AR(1) treated as a $\sigma \to \infty$ limit (large noise, no deterministic backbone). The intermediate regime in which both the deterministic skeleton and the noise contribute is governed by the function $f(\sigma)$.

7 Applications

The geometric invariants κ and K are not merely discriminatory; they admit immediate operational use as estimators and diagnostics. We sketch three applications. Each can be implemented in $\mathcal{O}(T)$ time on a single time series, and each is presented here at the level of an algorithmic recipe; detailed empirical validation on benchmark datasets is deferred to follow-up work.

7.1 Persistence classification via the hodograph

Setting. Given a stationary AR(1) Gaussian time series $\{y_t\}$ with unknown $\phi \in (-1, 1)$, estimate ϕ from T observations.

Recipe. Compute the kinematic embedding $\mathbf{q}$, the bivector trajectory $B(t) = \mathbf{q}(t-1) \wedge \mathbf{q}(t)$, and the empirical mean log-curvature $\hat{M} = \frac{1}{T-2} \sum_t \log \kappa(t)$. By Proposition 3.1, $\mathbb{E}[\log \kappa] \approx C_0 - \log(1-\phi^2)$ with $C_0 \approx -0.300$ universal under componentwise standardisation. Solving for ϕ:

$$\hat{\phi}^2_{\text{hodograph}} \;=\; 1 - \exp(C_0 - \hat{M}).$$

Comparison with classical estimators. The standard estimator of ϕ is the lag-1 autocorrelation $\hat{\rho}_1 = \widehat{\mathrm{Cov}}(y_t, y_{t-1}) / \widehat{\mathrm{Var}}(y_t)$. The hodograph estimator $\hat{\phi}^2_{\text{hodograph}}$ is its geometric analogue; the two should agree asymptotically and under model specification (Gaussian AR(1)) but diverge under model misspecification (heavy tails, non-linearity, GARCH-style heteroscedasticity), in which case the hodograph estimator captures persistence-like behaviour without committing to the AR(1) parametric form.

7.2 Regime-shift detection via rolling curvature

Setting. Given a long time series $\{y_t\}$, detect points $t^\star$ at which the underlying DGP changes (regime shift, structural break, phase transition).

Recipe. Compute $\kappa(t)$ and $K(t, \tau)$ on rolling windows of size W centred at each t. Track the rolling means $\hat{M}_W(t)$ and proportions $\hat{P}_W(t)$. A regime-shift candidate is a t at which both rolling statistics change abruptly beyond the within-regime sampling fluctuation. The combined test uses the joint $(\hat{M}_W, \hat{P}_W)$ as a two-dimensional diagnostic; abrupt motion in this plane flags a regime shift.

Connection to existing tools. CUSUM-style change-point detectors are conventionally applied to y_t or y_t^2 (level or variance breaks). The geometric invariants extend these to *curvature* breaks, sensitive to changes in the underlying joint distribution that are not detectable from level or variance alone. The example of Section 6.4 on GARCH versus AR persistence suggests such breaks exist and are operationally distinct.

7.3 Volatility-vs-level diagnostic via curvature sign

Setting. Given a time series with persistent behaviour, diagnose whether the persistence is in the level (AR style) or in the volatility (GARCH style).

Recipe. Compute the mean Gaussian curvature $\bar{K}$ on the scalar channel (§4) over a fixed lag window. By the empirical pattern of § 6:

- $\bar{K} > 0$: volatility-clustering regime (GARCH-like).
- $\bar{K} < 0$: level-persistence regime (AR-like, integrated).
- $\bar{K} \approx 0$: white noise or anti-persistent regime.

Comparison with classical tools. Distinguishing volatility clustering from level persistence is conventionally done with GARCH-LM tests (Engle 1982) on squared residuals, requiring a choice of pre-filter. The mean Gaussian curvature is a direct, unfiltered diagnostic that classifies persistence by mechanism without requiring AR parametric specification. Its $\mathcal{O}(T)$ computational cost matches the classical tools while requiring fewer specification choices.

8 Limitations and Open Questions

We have, in deliberate Hamilton-style fashion, declared a programme and exhibited its first applications. The honest scientific position is that the programme has limitations and that a number of open questions remain. We discuss each in turn.

8.1 Reparametrisation behaviour

The Theorema Egregium guarantees the invariance of the Gaussian curvature under *isometric* transformations of the surface; arbitrary reparametrisations of the time-axis or of the lag-axis are not isometric and will move $\Pr[K > 0]$ and similar functionals. Operationally, the smoke-test of § 6.3 establishes that the sensitivity to a 10% jitter of timestamps is one to two orders of magnitude smaller than the sensitivity to DGP, so the invariants are robust enough for practical work. The honest theoretical position, however, is that the curvature is intrinsic to the discretised series *at a fixed sampling rate*, not to the underlying continuous-time process. In particular:

- The hodograph $\mathbb{E}[\log \kappa]$ depends on the discretisation step Δt: rescaling time changes the curvature by Δt^{-2} asymptotically.
- The Egregium $\Pr[K > 0]$ depends on the joint scaling of t and τ: the surface shape depends on the relative units chosen for the two parameters.

A genuinely parametrisation-invariant version would require constructing the corresponding continuous-time process limit and working with limit invariants, which is beyond the scope of the present paper.

8.2 Window dependence

For the Egregium reading the maximum lag $T_{\max}$ is a hyperparameter. The reported smoke tests use $T_{\max} = 15$ for bivector and $T_{\max,1} = 7$ for trivector channels. A systematic sensitivity analysis across $T_{\max}$ is pending; preliminary inspection suggests that $T_{\max} \in [5, 20]$ gives qualitatively stable discrimination, with the absolute t-statistic increasing with $T_{\max}$ up to a saturation point determined by the sample size T.

For the hodograph, the only window choice is the lag-1 wedge $\mathbf{q}(t-1) \wedge \mathbf{q}(t)$ used here; longer-lag wedges $\mathbf{q}(t-\tau) \wedge \mathbf{q}(t)$ give parallel constructions whose discrimination profile differs from the lag-1 case. We have not explored these.

8.3 Higher-order curvature analogues

The Frenet-Serret apparatus on a 1-curve in $\mathbb{R}^3$ assigns two invariants: curvature κ and torsion τ. We have used κ in this paper; τ remains unexploited. Analogously, on a 2-surface in $\mathbb{R}^3$ the Gauss-Bonnet apparatus gives K (Gaussian) and H (mean) curvatures plus their integrated forms; we have used K, leaving H and the integrated quantities unexploited. A complete geometric profile of a kinematic embedding would include all of these as separate diagnostic channels. The present paper is a first probe of the geometric content; the exhaustive treatment is deferred.

The trivector channel has been touched (Egregium, three-window products) but not developed at the same level as the bivector. The natural extension is a full 3-D Frenet-Serret on a 3-submanifold of $\mathbb{R}^4$, with the corresponding sectional curvatures and the scalar curvature as invariants; this is, in principle, mechanical from the present apparatus but is left as future work.

8.4 Theoretical closed forms

Proposition 3.1 is, as stated, conjectural with empirical support to better than 0.01 at $\phi = 0.9$. A rigorous derivation requires computing the joint distribution of $V(t)$ and $A(t)$ as a function of ϕ and evaluating $\mathbb{E}[\log(|V \times A|/|V|^3)]$. The cube in the denominator prevents direct application of Wick's theorem, but an asymptotic expansion in $1 - \phi^2$ via the moment generating function of the Gaussian field is available. The full derivation is the natural target of a companion technical note.

For the Egregium reading, the closed form $\Pr[K > 0]$ for canonical DGPs is, to our knowledge, an open analytical question. The smoke-test discrimination patterns suggest a closed form exists for AR(1) (monotone in ϕ^2) and for fBm (monotone in H), but rigorous statements await.

8.5 Embedding choices

The kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ is the discrete analogue of $(y, \dot{y}, \ddot{y})$ and is the default in the GSD programme by reference to Newton's kinematic state. Other choices are possible: the lag-embedding basis (y_t, y_{t-1}, y_{t-2}) is Pascal-equivalent (Lemma 5.2 of Paper 27); polynomial-basis embeddings, exponential-decay embeddings, and signature-based embeddings (Lyons, 1998) are alternatives. Whether the geometric invariants of this paper transfer cleanly across embedding choices is open. The Hamilton-1847 hodograph construction was about velocity specifically; whether the GSD hodograph reading depends on the kinematic embedding being a *kinematic* embedding (rather than any 3-vector embedding) is worth investigating.

9 Closing

9.1 Two ancestors, one foundation

This paper is, in its essential argument, an instantiation of two 19th-century results on 21st-century data. Hamilton's 1847 hodograph theorem—that the curvature of the velocity locus is constant under inverse-square gravity—asserted that a deterministic dynamical trajectory carries an intrinsic geometric fingerprint of the underlying force law. Gauss's 1827 Theorema Egregium—that the Gaussian curvature is intrinsic to a surface— asserted that a deterministic geometric object carries information independent of the chosen ambient embedding. Both results are exact statements about deterministic dynamics. They invite the question of what geometric content adheres to an empirical signal that combines a deterministic skeleton with stochastic perturbation.

The kinematic embedding $\mathbf{q}(t) \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)$ supplies a canonical answer in the manner of mathematical physics: an *exact* theory on the deterministic skeleton, with stochastic perturbation handled *asymptotically* as $\sigma \to 0$, and detectability thresholds $\sigma^\star$ specific to each process. The two complementary curvatures—Frenet-Serret on a 1-curve, Gauss on a 2-surface—are constructible from the same embedding in $\mathcal{O}(T)$ time, computed exactly for deterministic trajectories (sine, polynomial, logistic chaos, Hénon) and empirically validated as DGP discriminators on stochastic processes with t-statistics up to $+24$ and $+91$ respectively. The framework, in this sense, is to time series geometry what Newtonian mechanics is to classical dynamics: exact on the underlying deterministic ground, asymptotically extensible to perturbations, and quantified by process-specific characteristic constants (the detectability thresholds).

9.2 Companion to Paper 28

The present paper is the geometric-differential face of a two-paper foundation. The companion is Paper 28 (Vázquez Broquá 2026b), which develops the parallel algebraic-probabilistic face: Hamilton's 1843 *Principle of Fluctuation* is instantiated as a non-abelian generalisation of the Cramér-Kolmogorov characteristic function, valued in $\mathrm{Spin}(3) \cong SU(2)$ and stratified by the irreducible representations of that group. The two papers share the algebraic substrate ($\mathrm{Cl}(3,0)$) and the conceptual move (replace abelian- or Cartesian-implicit foundations by their non-abelian or multivectorial extensions); they differ in the modality of geometry versus probability.

The parallelism is deliberate. Time series possess a geometry that the Cartesian timeline lacks; probability possesses a non- abelian extension that Cramér's 1942 abelianisation foreclosed. Both extensions are foundations, not corrections, and both recover the classical theory as projection. The further papers of the GSD programme—the spectral identity, the entropy decomposition, the graded transfer entropy, the moment tensor— all rest on these two foundations as derived consequences.

9.3 The argument is not yet complete

We have stated two empirical propositions and two theorems; we have not proved Proposition 3.1 in closed form. We have presented three applications at the level of recipes; we have not validated them on benchmark datasets. We have indicated higher-order curvature analogues; we have not exploited them. A complete treatment of the geometric foundation of GSD would close each of these open ends, and the present paper is therefore best read as a programmatic announcement, in Hamilton's manner, rather than as the final word.

9.4 Open programme

We close with three open directions:

1. *Closed-form derivation of Proposition 3.1.* The asymptotic expansion of $\mathbb{E}[\log \kappa]$ in $1 - \phi^2$ via the joint distribution of V and A for AR(1) Gaussian processes. Rigorous version of the empirical scaling.
2. *Application to benchmarks.* The three applications of § 7 should be validated on standard time-series benchmark datasets (CMAPSS, FRED-MD, financial returns). The persistence classifier, regime-shift detector, and volatility-vs-level diagnostic should be compared head-to-head with their classical counterparts (ACF, CUSUM, GARCH-LM).
3. *Continuum limits.* The discrete-time invariants of this paper have natural continuous-time analogues. For Brownian motion as the high-frequency limit of fBm $H = 0.5$, what is the limit law of $\mathbb{E}[\log \kappa]$ and $\Pr[K > 0]$? The answers, if they exist as standard probabilistic objects, would consolidate the connection to rough-path theory (Lyons 1998) and would close the genealogical loop with continuous-time stochastic geometry.

The geometry of a discrete time series, this paper has argued, is the kinematic embedding's. The claim is, in our reading, neither small nor self-evident. Whether it proves productive—whether the two-paper foundation finds operative purchase beyond the GSD programme that motivated it—is a question the readers of this note will collectively decide.

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